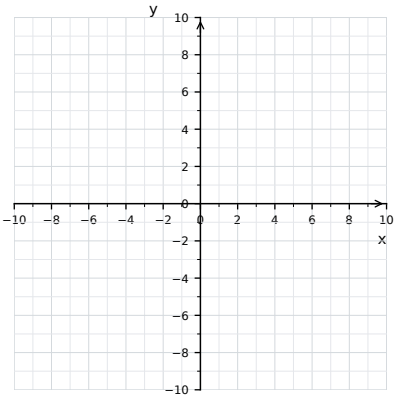
TSM 099 Final Exam

Practice 3 | Version 202610122020.33

Name: _____ Date: _____

Complete and show all work on separate paper. Write exact final answers at right.

QUESTIONS	ANSWERS
1. Solve by factoring. $x^2 + 2x = 24$	1. Solutions 10 pts
2. Add and simplify. $\frac{2}{x^2 - 4x} + \frac{3}{x^2 - 16}$	2. Expression 10 pts
3. Find the slope-intercept equation through these two points. $(2,\ 3);\ (4,\ 6)$	3. Equation 10 pts
4. Solve using the quadratic formula. Give exact simplified solutions. $4x^2 = 6 - x$	4. Solutions 10 pts
5. Factor completely. $12x^2 + 5x - 3$	5. Factors 10 pts
6. Evaluate at x = -2; show substitution. $(x + 3)(4x - 12)$	6. Value 10 pts
7. Find the intercepts and graph the line on the grid. $3x + 2y = 12$	7. Intercepts 4 pts Graph  6 pts
8. Solve by taking square roots. $3(x - 3)^2 = 75$	8. Solutions 10 pts

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QUESTIONS	ANSWERS
9. Multiply and simplify. $\frac{x^2-16}{x^2-2x-63} \times \frac{x+7}{x^2+4x}$	9. Expression 10 pts
10. Solve and check in the original equation. $\frac{12}{2x-10} + \frac{2}{x-5} = \frac{8}{3}$	10. Solution/check 10 pts

Instructor answer key: TSM 099 Final Exam

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Each column matches the vertical answer positions on its corresponding test page.

Concise grading reference; required work and checks are assessed on separate paper.

TEST PAGE 1

1. Solutions

$x = -6,\ 4$

10 pts

2. Expression

$$\frac{5x+8}{x(x-4)(x+4)}$$

10 pts

3. Equation

$y = \frac{3x}{2}$

10 pts

4. Solutions

$x = -\frac{1}{8} + \frac{\sqrt{97}}{8},\ -\frac{\sqrt{97}}{8} - \frac{1}{8}$

10 pts

5. Factors

$(3x-1)(4x+3)$

10 pts

6. Value

-20

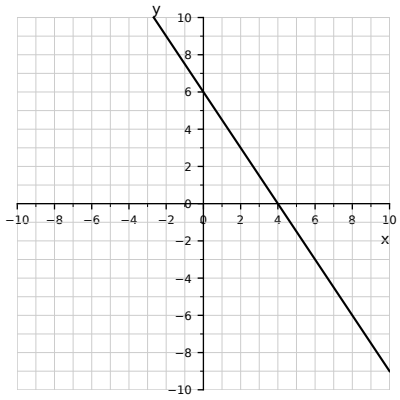
10 pts

7. Intercepts

$(4,\ 0);\ (0,\ 6)$

4 pts

Graph



6 pts

8. Solutions

$x = -2,\ 8$

10 pts

TEST PAGE 2

9. Expression

$$\frac{x-4}{x(x-9)}$$

10 pts

10. Solution/check

$x = 8$

10 pts